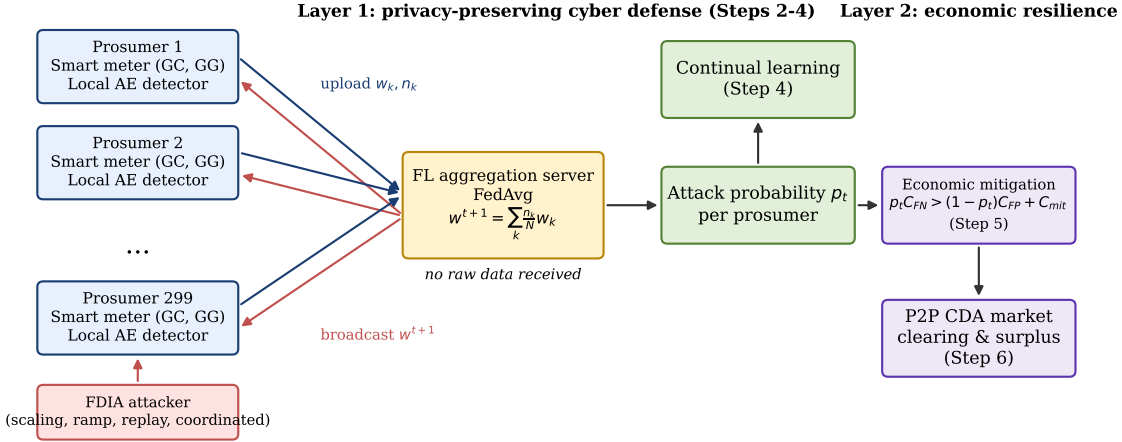
Layer 1: privacy-preserving cyber defense (Steps 2-4)
Layer 2: economic resilience
Prosumer 1
Smart meter (GC, GG)
Local AE detector
Prosumer 2
Smart meter (GC, GG)
Local AE detector
...
Prosumer 299
Smart meter (GC, GG)
Local AE detector
FDIA attacker
(scaling, ramp, replay, coordinated)
upload $w_k, n_k$
broadcast $w^{t+1}$
FL aggregation server
FedAvg
$w^{t+1} = \sum_k \frac{n_k}{N} w_k$
no raw data received
Continual learning
(Step 4)
Attack probability $p_t$
per prosumer
Economic mitigation
$p_t C_{FN} > (1 - p_t) C_{FP} + C_{mit}$
(Step 5)
P2P CDA market
clearing & surplus
(Step 6)